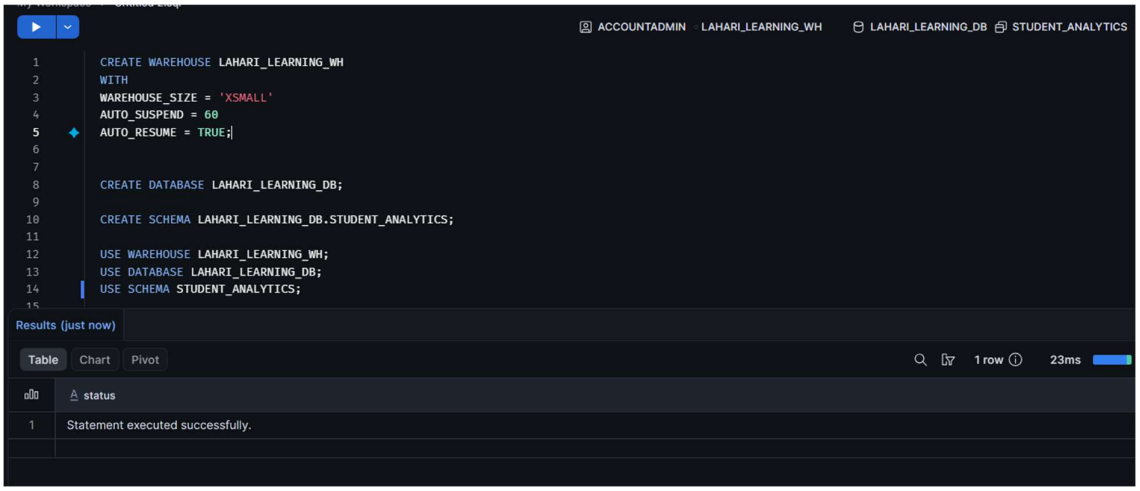
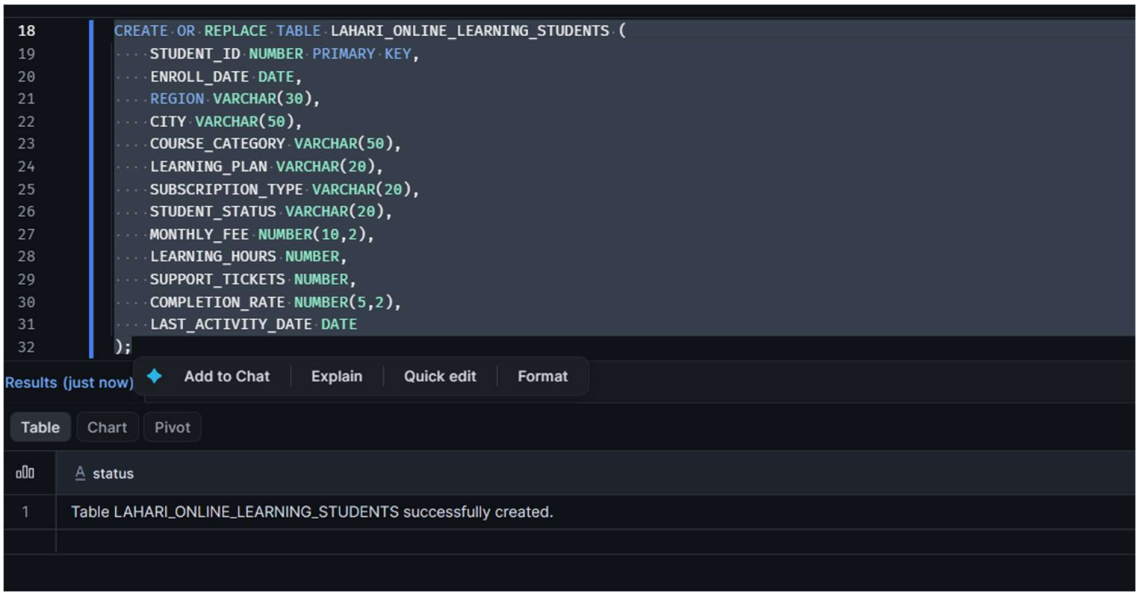
Online Learning Platform – Student Engagement & Subscription Analysis

Business scenario: An online learning platform provides programming, cloud computing, cybersecurity, data science and AI courses. The company wants to understand student engagement, subscription revenue, active students, course popularity and student churn.

Create database, schema and warehouse



Create Table



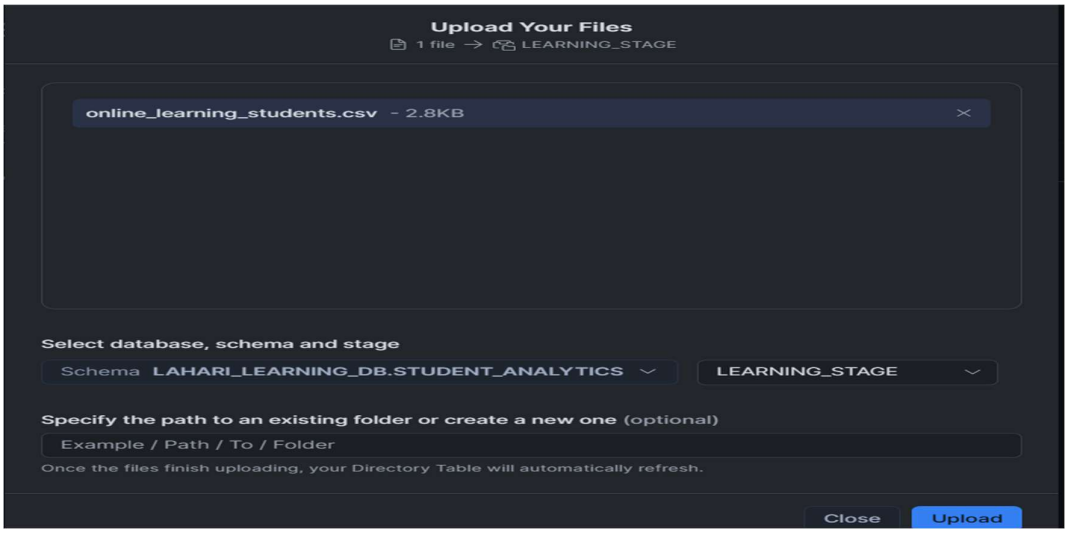
Uploading CSV file into Snowflake

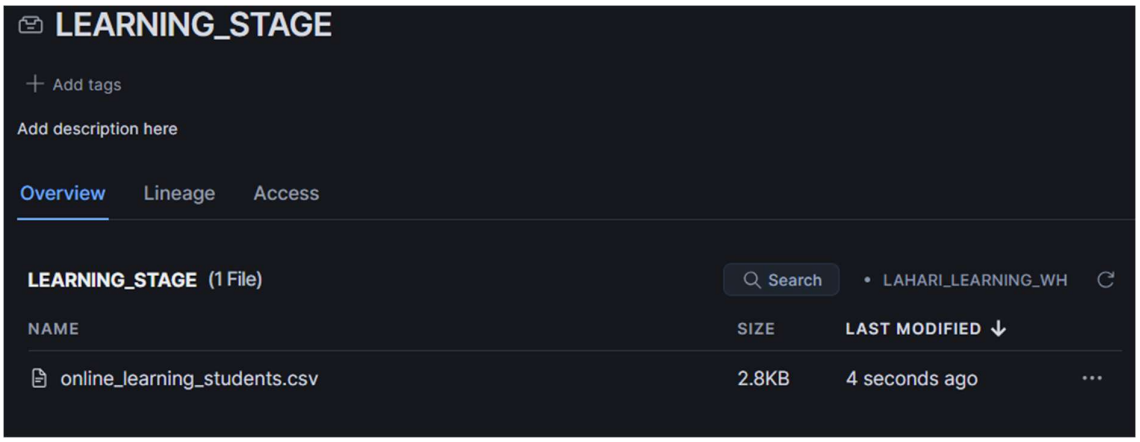
```
34  CREATE OR REPLACE FILE FORMAT LEARNING_CSV_FORMAT
35  TYPE = CSV
36  SKIP_HEADER = 1
37  FIELD_OPTIONALLY_ENCLOSED_BY = '"'
38  DATE_FORMAT = 'YYYY-MM-DD';
39
40  CREATE OR REPLACE STAGE LEARNING_STAGE
41  FILE_FORMAT = LEARNING_CSV_FORMAT;
```

esults (just now)

Table | Chart | Pivot

	status
1	Stage area LEARNING_STAGE successfully created.





```
COPY INTO LAHARI_ONLINE_LEARNING_STUDENTS
FROM @LEARNING_STAGE
FILE_FORMAT = (FORMAT_NAME = LEARNING_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';
```

Results (just now)

Table | Chart | Pivot — 1 row — 596ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	firs
1	learning_stage/online_l	LOADED	30	30	1	0	null	null	

Verify

```
SELECT * FROM LAHARI_ONLINE_LEARNING_STUDENTS;
```

Results (just now)

Table | Chart | Pivot — 30 rows — 118ms

	STUDENT_ID	ENROLL_DATE	REGION	CITY	COURSE_CATEGORY	LEARNING_PLAN	SUBSCRIPTION_TYPE	STUDENT_STATUS	MONTHL
	1001 1030	1/5/20255/22/2...	So... 30.0% West 26.7% +2 more	Beng... 10.0% Bhub... 10.0% +9 more	Programming 23.3% Cloud Computing 20.0% +3 more	Premium 36.7% Standard 36.7% +1 more	Monthly 60.0% Yearly 40.0%	Active 70.0% Churned 30.0%	399
1	1001	2025-01-05	South	Hyderabad	Data Science	Premium	Monthly	Active	
2	1002	2025-01-12	North	Delhi	Programming	Basic	Monthly	Churned	
3	1003	2025-02-03	West	Mumbai	Cloud Computing	Premium	Yearly	Active	
4	1004	2025-02-18	South	Chennai	Cybersecurity	Standard	Monthly	Active	
5	1005	2025-03-01	East	Kolkata	AI & ML	Premium	Yearly	Churned	
6	1006	2025-03-15	West	Pune	Programming	Standard	Yearly	Active	
7	1007	2025-04-07	South	Bengaluru	Data Science	Basic	Monthly	Active	

Total Number of Records

```
SELECT COUNT(*)
FROM LAHARI_ONLINE_LEARNING_STUDENTS;
```

Results (just now)

Table | Chart | Pivot

	COUNT(*)
1	30

There are no duplicate student ids and null values present in the dataset

Check student status

```
SELECT STUDENT_STATUS, COUNT(*)
FROM LAHARI_ONLINE_LEARNING_STUDENTS
GROUP BY STUDENT_STATUS;
```

Results (2 minutes ago)

Table | Chart | Pivot

	STUDENT_STATUS	COU
1	Active	21
2	Churned	9

Revenue by course

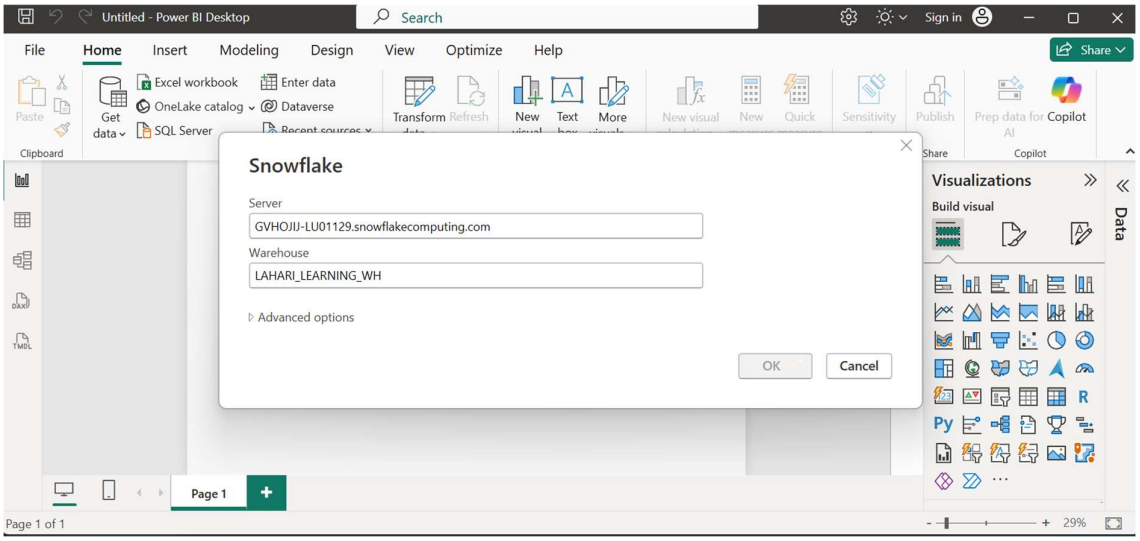
```sql
SELECT
    COURSE_CATEGORY,
    SUM(MONTHLY_FEE) AS MONTHLY_REVENUE
FROM LAHARI_ONLINE_LEARNING_STUDENTS
WHERE STUDENT_STATUS = 'Active'
GROUP BY COURSE_CATEGORY
ORDER BY MONTHLY_REVENUE DESC;
```

Results (just now)

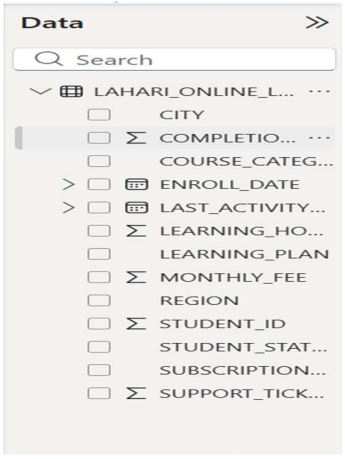
Table | Chart | Pivot — 5 rows

	COURSE_CATEGORY	MONTHLY_REVENUE
1	Cloud Computing	4645.00
2	Programming	3395.00
3	Data Science	2796.00
4	AI & ML	2647.00
5	Cybersecurity	2296.00

Connect to Power BI

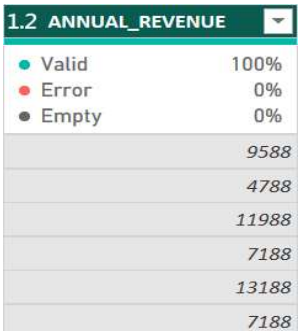


Data Loaded

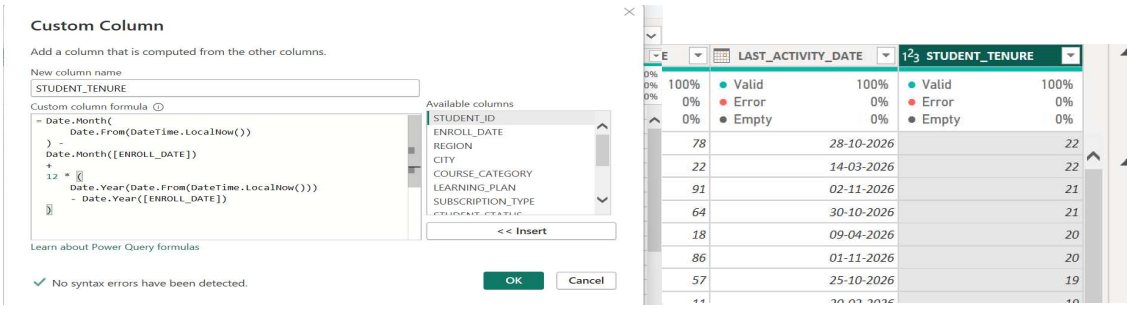


Adding New Columns

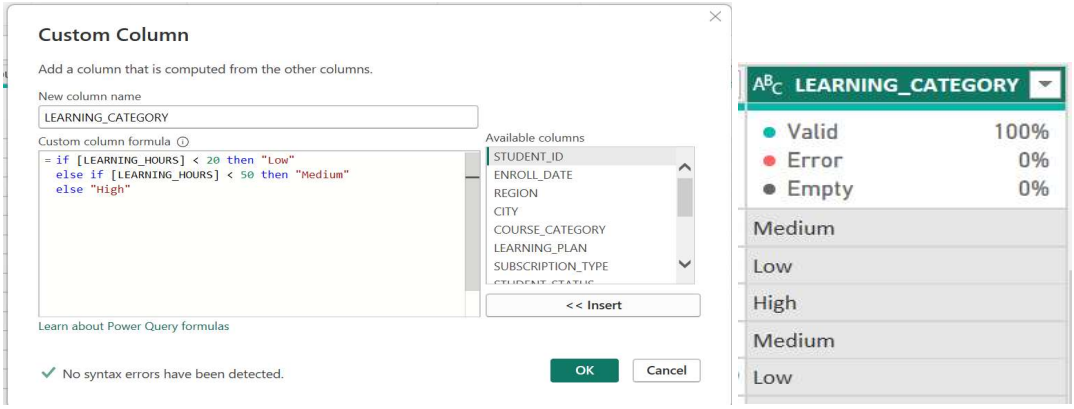
1. Annual_Revenue: Estimates annualized subscription revenue.



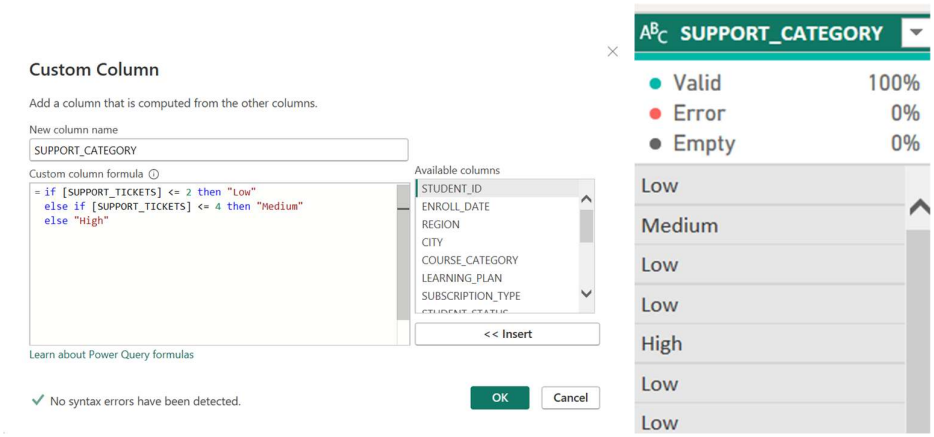
2. Student_Tenure: Calculates the approximate number of months since enrollment.



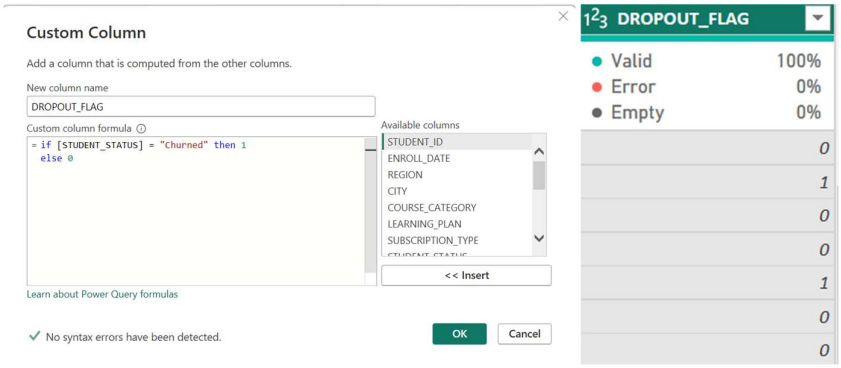
3. Learning_category: Classifies students according to learning hours.



4. Support_category: Classifies students based on support requests.

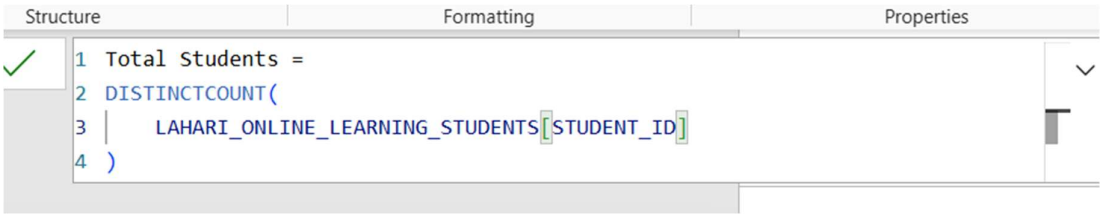


5. Dropout_flag: Converts student status into a numerical value.

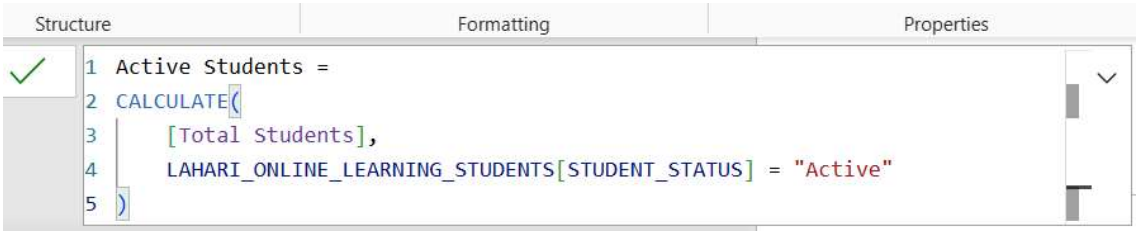


Creating DAX Measures

1. Total Students: Counts Unique students



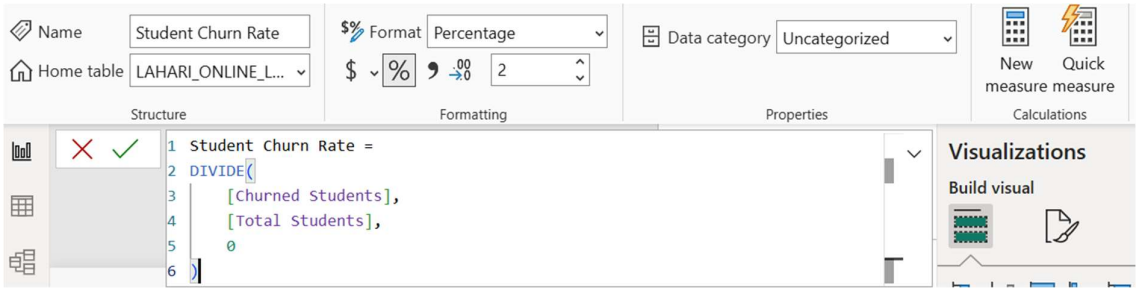
2. Active Students: Counts currently active students.



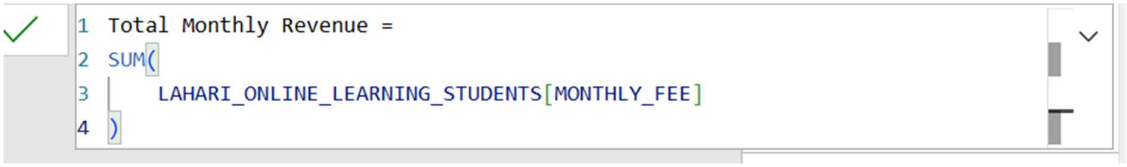
3. Churned Students: Counts students who discontinued their subscriptions.



4. Student Churn Rate



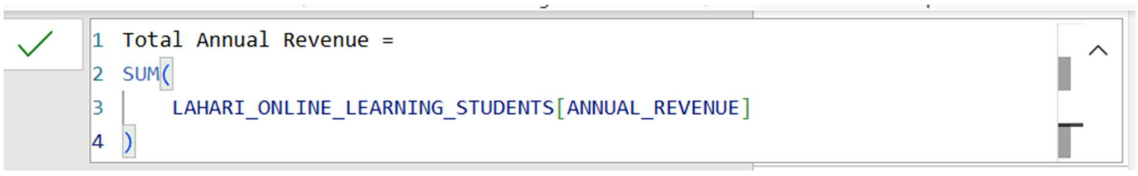
5. Total Monthly Revenue: Calculates monthly subscription revenue represented by the dataset.



6. Average Monthly Revenue: Calculates average monthly subscription fee per student.



7. Total Annual Revenue: Calculates estimated annualized subscription revenue.



8. Revenue Lost Due to Churn: Calculates the monthly subscription value associated with churned students. This is estimated revenue at risk, not verified historical revenue loss.

```
Revenue Lost Due to Churn =
CALCULATE(
    [Total Monthly Revenue],
    LAHARI_ONLINE_LEARNING_STUDENTS[STUDENT_STATUS] = "Churned"
)
```

9. Average Learning Hours

```
Average Learning Hours =
AVERAGE(
    LAHARI_ONLINE_LEARNING_STUDENTS[LEARNING_HOURS]
)
```

10. Average Completion Rate

```
Average Completion Rate =
AVERAGE(
    LAHARI_ONLINE_LEARNING_STUDENTS[COMPLETION_RATE]
)
```

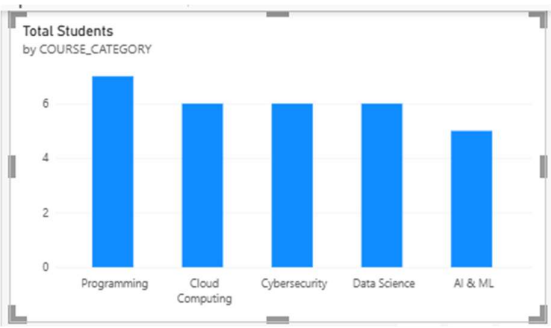
Visualizations

Visualization 1: Student Count by Course Category

Business insight: Identifies which course categories have the highest student enrollment.

Interpretation: Shows whether Data Science, Programming, Cloud Computing, Cybersecurity or AI & ML attracts more students.

Recommendation: Focus course development and marketing on categories with high enrollment and investigate opportunities in less popular categories.

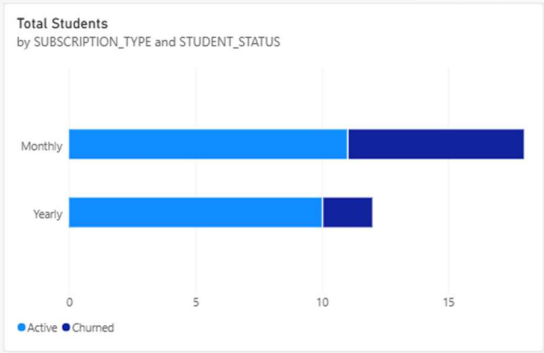


Visualization 2: Student Churn by Subscription Type

Business insight: Compares Active and Churned students across Monthly and Yearly subscriptions.

Interpretation: Identifies which subscription type has a larger proportion of churned students.

Recommendation: Provide retention offers, course discounts or additional learning support for subscription categories with higher churn.

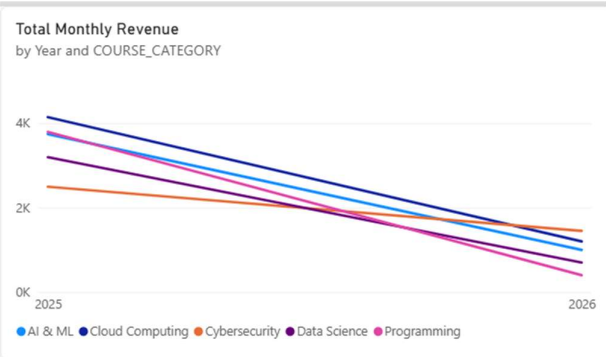


Visualization 3: Monthly Revenue by Course Category

Business insight: Compares subscription revenue across different course categories over enrollment months.

Interpretation: Identifies which course categories contribute more subscription revenue across the recorded enrollment periods.

Recommendation: Focus on categories generating higher subscription value and investigate changes in enrollment patterns.

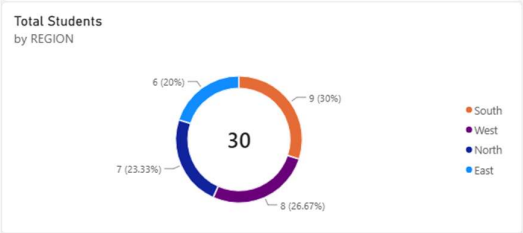


Visualization 4: Student Distribution by Region

Business insight: Shows the geographical distribution of students.

Interpretation: Identifies regions with the highest and lowest student enrollment.

Recommendation: Focus marketing campaigns and regional learning programs on locations with growth potential.

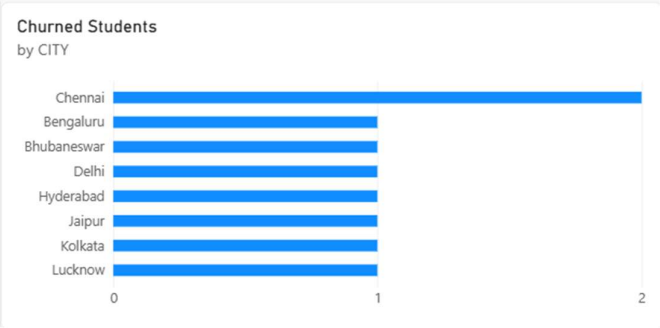


Visualization 5: Churned Students by City

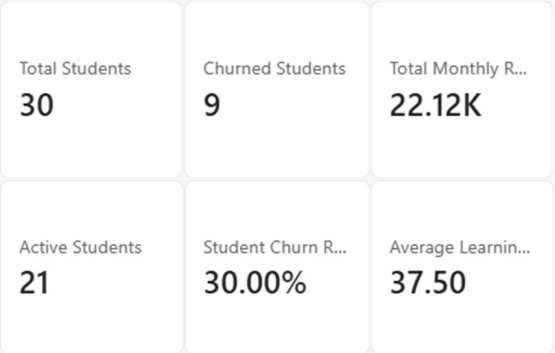
Business insight: Identifies cities with the highest number of discontinued students.

Interpretation: Highlights geographical areas where student retention needs attention.

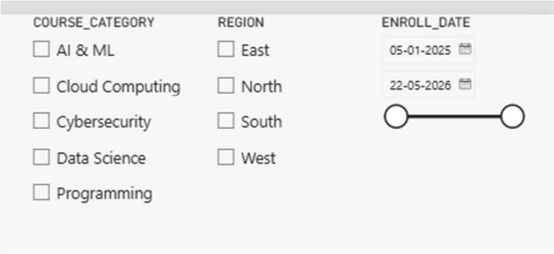
Recommendation: Investigate course quality, pricing, student engagement and support services in cities with higher churn.



Visualization 6: Card Visuals



Visualization 7: Slicers



Final Dashboard:

